

NEET Biology MCQ Sample

30 original, reviewed MCQs with answers and NCERT-aligned explanations

How to use this sample

Attempt each question without checking the answer. Mark uncertain items, then use the answer section to identify the NCERT concept behind each mistake. This sample contains only MedQGo original practice questions; it does not reproduce an official question paper.

Error-review sheet

For every incorrect answer, write: (1) chapter, (2) exact NCERT concept, (3) why your chosen option looked plausible, and (4) one recall cue for the next attempt.

Questions

1. In animal tissues, the important intermediate product from glucose for synthesizing fatty acids is

- A. pyruvate
- B. UTP
- C. acetyl-CoA
- D. acetone acid

NCERT Biology Class 11, Chapter 7

2. A couple with no phenotypic symptoms of phenylketonuria have given birth to three children affected by the condition. What is the probability that they will have a phenotypically normal child in their future offspring?

- A. 25%
- B. 50%
- C. 75%
- D. 100%

NCERT Biology Class 12, Chapter 7

3. Which of the following does not belong to the internal environment?

- A. cellular fluid
- B. interstitial fluid
- C. lymph
- D. cerebrospinal fluid

NCERT Biology Class 11, Chapter 15

4. The main chemical bond in protein molecules is

- A. peptide bonds
- B. salt bridges
- C. ester bonds
- D. disulfide bonds

NCERT Biology Class 11, Chapter 9

5. Which of the following is the main site for biological oxidation and energy conversion in animal cells?

- A. nucleus
- B. endoplasmic reticulum
- C. golgi apparatus
- D. mitochondria

NCERT Biology Class 11, Chapter 8

6. If an extra chromosome is added to a diploid cell, it can form

- A. triploid
- B. monoploid
- C. monosome
- D. trisomy

NCERT Biology Class 12, Chapter 4

7. Which of the following components in bacteria is extrachromosomal genetic material?

- A. plasmid
- B. rRNA
- C. mRNA
- D. metachromatic granules

NCERT Biology Class 12, Chapter 5

8. The disease in children due to iodine deficiency or hypothyroidism during embryonic development is

- A. Addison's disease
- B. Dwarfism
- C. Acromegaly
- D. Cretinism

NCERT Biology Class 11, Chapter 19

9. Which of the following conditions can lead to protein in urine?

- A. increase in glomerular filtration rate
- B. reduction in slit diaphragm number on podocytes
- C. increase in filtration fraction
- D. acceleration of tubular fluid flow

NCERT Biology Class 11, Chapter 16

10. The basic structural unit of eukaryotic chromatin is

- A. alpha-helix
- B. nucleosome
- C. plasmid
- D. beta-sheet

NCERT Biology Class 11, Chapter 2

11. The strategy adopted by the Human Genome Project for sequencing is

- A. mapping followed by sequencing and shotgun sequencing combined
- B. mapping followed by sequencing
- C. shotgun sequencing

D. chip sequencing

NCERT Biology Class 12, Chapter 5

12. If a person's menstrual cycle is between 28 to 30 days, during which days of the cycle would ovulation most likely occur?

- A. 6th and 8th day
- B. 10th and 12th day
- C. 14th and 16th day
- D. 18th and 20th day

NCERT Biology Class 12, Chapter 2

13. The oxygen dissociation curve represents the relationship between

- A. oxygen content and hemoglobin concentration
- B. hemoglobin oxygen saturation and blood oxygen pressure
- C. hemoglobin oxygen saturation and hemoglobin oxygen content
- D. hemoglobin concentration and hemoglobin oxygen capacity

NCERT Biology Class 11, Chapter 14

14. Which of the following is a function of macrophages in immunology?

- A. secrete specific antibodies
- B. present antigens
- C. mediate type III hypersensitivity reaction
- D. mediate type I hypersensitivity reaction

NCERT Biology Class 12, Chapter 7

15. In a population, the frequency of the genotype AA at a single nucleotide polymorphism (SNP) locus is 64%, AG is 32%, and GG is 4%. What is the frequency of the allele A?

- A. 0.64
- B. 0.16
- C. 0.9
- D. 0.8

NCERT Biology Class 12, Chapter 4

16. During meiosis II, separation occurs of

- A. sister chromatids
- B. non-sister chromatids
- C. homologous chromosomes
- D. alleles

NCERT Biology Class 11, Chapter 10

17. Polyacrylamide gel electrophoresis has higher resolution than general electrophoresis due to its

- A. gradient concentration effect
- B. molecular sieving effect
- C. charge aggregation effect
- D. viscosity effect

NCERT Biology Class 12, Chapter 9

18. In striated muscle, the initiating step for myofilament sliding is

- A. calcium ions in the sarcoplasm bind to troponin
- B. myosin ATPase activity increases leading to ATP breakdown
- C. myosin binds to actin
- D. cross-bridge binds to actin

NCERT Biology Class 11, Chapter 17

19. Due to a mutation causing a codon to form a stop codon, this mutation is known as

- A. missense mutation
- B. nonsense mutation
- C. stop codon mutation
- D. frameshift mutation

NCERT Biology Class 12, Chapter 5

20. During the depolarization phase of an action potential, the direct force promoting Na^+ influx is

- A. concentration gradient
- B. electrical gradient
- C. sodium pump activity
- D. ATP breakdown

NCERT Biology Class 11, Chapter 18

21. Familial hypercholesterolemia is inherited through

- A. autosomal recessive
- B. autosomal dominant
- C. X-linked recessive
- D. X-linked dominant

NCERT Biology Class 12, Chapter 4

22. In the citric acid cycle, which high-energy compound is directly generated through substrate-level phosphorylation?

- A. TTP
- B. ATP
- C. GTP

D. UTP

NCERT Biology Class 11, Chapter 12

23. Which of the following populations is in genetic equilibrium?

- A. AA: 0.20; Aa: 0.60; aa: 0.20
- B. AA: 0.25; Aa: 0.50; aa: 0.25
- C. AA: 0.30; Aa: 0.50; aa: 0.20
- D. AA: 0.50; Aa: 0; aa: 0.50

NCERT Biology Class 12, Chapter 4

24. Which of the following cannot be used as a gene cloning vector?

- A. plasmid
- B. M13 bacteriophage
- C. viral RNA
- D. λ bacteriophage

NCERT Biology Class 12, Chapter 9

25. The spatial conformation of proteins mainly depends on

- A. the sequence of amino acids in the polypeptide chain
- B. α -helix and β -sheet
- C. side chains of amino acids in the polypeptide chain
- D. positions of disulfide bonds in the polypeptide chain

NCERT Biology Class 11, Chapter 9

26. In the genetic code table, the genetic codons are represented by which of the following molecules' 5'→3' direction base triplets?

- A. DNA
- B. RNA
- C. tRNA
- D. rRNA

NCERT Biology Class 12, Chapter 5

27. What is the role of RNA primer during DNA replication?

- A. Provides a 3'-OH end for dNTP addition
- B. Extends Okazaki fragments
- C. Participates in forming the primosome
- D. Assists helicase action

NCERT Biology Class 12, Chapter 5

28. Where does aerobic respiration and oxidative phosphorylation occur in a cell?

- A. ribosome

- B. mitochondrion
- C. cell membrane
- D. rough endoplasmic reticulum

NCERT Biology Class 11, Chapter 8

29. In the genetic code table, the genetic codons are represented by which molecule's 5'→3' direction of base triplets?

- A. mRNA
- B. tRNA
- C. rRNA
- D. DNA

NCERT Biology Class 12, Chapter 5

30. Which of the following is an incorrect statement about mutations?

- A. Point mutations include transitions and transversions
- B. Insertion of 2 bp can cause a frameshift mutation
- C. Mutations can alter the genotype
- D. If there is a mutation, it will definitely affect the phenotype

NCERT Biology Class 12, Chapter 5

Answers and revision notes

1. Answer: C. acetyl-CoA

Acetyl-CoA is the key intermediate in the conversion of glucose to fatty acids.

2. Answer: C. 75%

Phenylketonuria (PKU) is an autosomal recessive disorder. If both parents are carriers, the probability of having a child affected by PKU is 25%, and the probability of having a phenotypically normal child is 75%. The couple has already had three affected children, which suggests they are likely carriers.

3. Answer: A. cellular fluid

The internal environment, also known as the extracellular environment, includes blood plasma, interstitial fluid, and lymph. Cerebrospinal fluid is primarily found within the central nervous system and does not directly participate in general body fluid exchange.

4. Answer: A. peptide bonds

Peptide bonds are the primary covalent bonds that link amino acids together to form polypeptide chains in proteins.

5. Answer: D. mitochondria

Mitochondria are known as the powerhouses of the cell, where most of the ATP (adenosine triphosphate) is generated through cellular respiration.

6. Answer: D. trisomy

Trisomy refers to the presence of an extra chromosome in a cell. In a diploid cell, adding one extra chromosome results in trisomy.

7. Answer: A. plasmid

The genetic material in bacteria includes chromosomes, plasmids, transposons, and prophages. Plasmids are extrachromosomal genetic materials found in the cytoplasm of bacterial cells as circular or linear dsDNA that can replicate independently. Plasmids are not essential for the life activities of bacteria but carry information that can confer specific biological traits to the host bacterium, such as conjugation ability, antibiotic resistance, and pathogenicity.

8. Answer: D. Cretinism

Addison's disease is due to adrenal cortex dysfunction. Dwarfism results from growth hormone deficiency or inadequate secretion. Acromegaly occurs due to excessive growth hormone in adults. Cretinism involves intellectual disability and developmental delay in children due to iodine deficiency or hypothyroidism during embryonic development.

9. Answer: B. reduction in slit diaphragm number on podocytes

Slit diaphragms are special proteins expressed on the foot processes of podocytes in the glomerulus. They play a crucial role in forming the filtration barrier and preventing blood proteins from being filtered into urine. Therefore, their reduction leads to proteinuria.

10. Answer: B. nucleosome

Nucleosomes are the fundamental repeating subunits of chromatin in eukaryotes, consisting of DNA wrapped around histone proteins.

11. Answer: A. mapping followed by sequencing and shotgun sequencing combined

The Human Genome Project used a combination of mapping followed by sequencing along with shotgun sequencing to achieve accurate genome sequencing.

12. Answer: C. 14th and 16th day

Ovulation typically occurs around 14 days before the next menstrual period. For a cycle between 28 to 30 days, ovulation would most likely occur between the 14th and 16th day.

13. Answer: B. hemoglobin oxygen saturation and blood oxygen pressure

The oxygen dissociation curve illustrates the relationship between hemoglobin oxygen saturation and blood oxygen pressure.

14. Answer: B. present antigens

Macrophages have the following functions: killing and removing pathogens, killing intracellular parasites and tumor cells, participating in inflammatory responses, processing and presenting antigens to initiate adaptive immune responses, and immunoregulatory activities.

15. Answer: D. 0.8

The frequency of the allele A can be calculated as $(2 * \text{frequency of AA} + \text{frequency of AG}) / 2 = (2 * 0.64 + 0.32) / 2 = 0.8$.

16. Answer: A. sister chromatids

During meiosis II, the sister chromatids separate and move to opposite poles.

17. Answer: B. molecular sieving effect

Polyacrylamide gel electrophoresis has higher resolution than general electrophoresis due to its molecular sieving effect, which allows separation based on the size of molecules.

18. Answer: A. calcium ions in the sarcoplasm bind to troponin

Calcium ions in the sarcoplasm bind to troponin, which is the initiating step for myofilament sliding.

19. Answer: B. nonsense mutation

A nonsense mutation results in the formation of a premature stop codon, leading to a truncated protein.

20. Answer: B. electrical gradient

The direct force promoting Na^+ influx during the depolarization phase of an action potential is the electrical gradient. The concentration gradient and sodium pump activity are important but not the direct driving force, while ATP breakdown provides energy but does not directly drive ion movement.

21. Answer: B. autosomal dominant

Familial hypercholesterolemia is typically an autosomal dominant disorder, meaning it can be inherited from one affected parent.

22. Answer: B. ATP

Substrate-level phosphorylation in the citric acid cycle does not produce ATP; it occurs during glycolysis and oxidative phosphorylation. The other options are nucleoside triphosphates but not directly generated through substrate-level phosphorylation in the citric acid cycle.

23. Answer: B. AA: 0.25; Aa: 0.50; aa: 0.25

A population is in genetic equilibrium if the frequencies of alleles and genotypes remain constant from one generation to another, as described by the Hardy-Weinberg principle. The given genotype frequencies for option B (AA: 0.25; Aa: 0.50; aa: 0.25) satisfy this condition.

24. Answer: C. viral RNA

Plasmids, M13 bacteriophage, and λ bacteriophage are commonly used as vectors in recombinant DNA technology. Viral RNA cannot be directly used as a cloning vector.

25. Answer: A. the sequence of amino acids in the polypeptide chain

Proteins are large biological molecules formed by many amino acids linked through peptide bonds. The primary structure of a protein refers to the sequence of amino acids from N-terminus to C-terminus, i.e., the amino acid sequence. Secondary structure refers to the local spatial arrangement of the main chain, mainly including α -helices and β -sheets. Tertiary structure involves the overall spatial arrangement of all atoms in the polypeptide chain and its side chains. Quaternary structure concerns the aggregation of protein subunits. The first three structures are collectively known as the spatial structure of proteins. The primary structure is the foundation for spatial conformation, meaning that the spatial conformation of proteins mainly depends on the sequence of amino acids in the polypeptide chain.

26. Answer: B. RNA

The genetic code is read from the 5' to 3' direction of mRNA during protein synthesis.

27. Answer: A. Provides a 3'-OH end for dNTP addition

DNA polymerase cannot catalyze the formation of a phosphodiester bond between two free dNTPs; it can only catalyze the addition of dNTPs to the 3'-OH end of a nucleotide fragment. The primase, which is an RNA polymerase, provides a short RNA primer that offers a 3'-OH end for DNA synthesis. In the presence of DNA polymerase, dNTPs are added one by one to form the DNA daughter strand. B: Extending Okazaki fragments is done by DNA pol I. C: Primase participates in forming the primosome; after catalyzing primer formation, it enters the phase of DNA daughter strand elongation. D: The action of assisting helicase is performed by DnaC protein.

28. Answer: B. mitochondrion

Aerobic respiration and oxidative phosphorylation take place in the mitochondria of a cell.

29. Answer: A. mRNA

The genetic code is read from the mRNA during protein synthesis. The codons in the mRNA are a sequence of three nucleotides that specify which amino acid will be added next to the growing polypeptide chain.

30. Answer: D. If there is a mutation, it will definitely affect the phenotype

Mutations do not necessarily lead to changes in the phenotype; they may only alter the genotype.